

ANKIT KUMAR

Senior Software Engineer

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SUMMARY

Senior Software Engineer building scalable, high-throughput backend systems in Java and Spring Boot — spanning distributed logistics, SSO/IAM, payments, loyalty, and real-time communication. Owns systems processing 400K+ daily orders, 100,000+ daily payment users, and 50K+ real-time users, backed by Kafka, Redis, Cassandra, and AWS. Recent wins include a 40% downtime reduction and ₹1.3 Cr (~\$155K) in annual telecom savings.

SKILLS

Backend, Cloud & Distributed Systems: Java, Spring Boot, Spring Security, Apache Kafka, Redis, SQL / MySQL, AWS, Distributed Systems, System Design, Microservices Architecture
Data & Messaging Infrastructure: Docker, RabbitMQ, Elasticsearch, MongoDB, Cassandra
Architecture & Engineering Practices: REST APIs, Event-Driven Architecture, Multithreading & Concurrency, Circuit Breaker & Resilience Patterns, Design Patterns, Data Structures & Algorithms
Security: OAuth 2.0, JWT & SSO/LDAP
Tools: Git, Maven, Jenkins, JUnit & Mockito
AI-Assisted Development: Claude Code, Cursor, GitHub Copilot

EXPERIENCE

Senior Software Engineer 06/2025 – Present Noida, U.P.
Jubilant FoodWorks Limited

- Delivery Management System** | Java · Spring Boot · Kafka · Cassandra · MQTT · Redis · MySQL
- Owned end-to-end design and delivery of the DMS — rider allocation, multi-order clubbing, payout, and the incentive engine — supporting **400K+ daily orders** across **25K+ riders** and **2,500+ stores**, held to an enforced **80% test-coverage gate**.
 - Core contributor to a 14-module logistics engine ingesting ~58 GPS updates/sec over Cassandra, Kafka, and MQTT; built allocation as a chain of **9 rule classes** with anti-spoof geofencing and dual routing providers (OSRM primary, Google Routes fallback) behind a **circuit breaker**, cutting downtime by **40%**.
- Single Sign On & IAM Platform** | Java · Spring Boot · Spring Security · OAuth 2.0 · JWT · LDAP · Caffeine
- Designed and shipped a centralized SSO/IAM service supporting OAuth 2.0, JWT, LDAP, and API-key access, with multi-tenant SSO spanning **120+ tenant-aware code paths** that replaced duplicated per-service auth logic.
 - Established a server-to-server (S2S) auth library, trimming integration effort for backend teams by **80%**; mentored junior engineers and drove **code review standards** across the backend team in an Agile setup.

- Delivery Time Prediction (ML)** | Java · Spring Boot · ONNX Runtime · XGBoost · AWS S3
- Built an ONNX Runtime inference service around a multi-output regression model splitting prediction into prep, transit, and last-mile segments, reaching **85-95% per-segment accuracy** with batch prediction at **500 orders per call**; output feeds the customer-facing delivery promise and SLA reporting.

Software Engineer 07/2022 – 06/2025 Noida, U.P.
Jubilant FoodWorks Limited

- Loyalty Rewards Service** | Java · Spring Boot · Kafka · Redis · Liquibase · MySQL
- Led development of a card-based points ledger where transfers redeem the source card and mint a new one, wrapped in **optimistic-lock retry** to survive concurrent wallet writes; lifted customer engagement by **30%** and repeat business by **35%**.
 - Built **Kafka-driven** async point issuance and batched card-expiry processing off the request path, decoupling reward crediting from checkout latency at nationwide order volume, with **Redis** fronting the busiest wallet and milestone lookups.
- Data Call Service** | Java · Spring Boot · WebRTC · STUN/TURN · FCM
- Built the WebRTC signalling layer on **Netty-SocketIO** (offer/answer/ICE relay, HMAC-token validation, full call-state logging) with an Opus-only, video-disabled **32 kbps** profile for constrained mobile data — slashing telecom costs by **60%** and saving **INR 1.3 crore (~USD 155K)** annually for **50K+ users**.
- Job Scheduling Service** | Java · Spring Boot · Quartz · Redis · MySQL
- Built three execution modes off one dispatcher and made terminal-state writes race-proof with an **atomic conditional update**, so a late callback and the timeout checker can never both resolve a job; tiered Redis caching cut repeat database reads on dashboards by an estimated **60-70%**.
- Unified Payment Service** | Java · Spring Boot · MySQL · Elasticsearch · RabbitMQ
- Built a multi-PSP orchestrator (**PayU, Paytm, Razorpay**) as pluggable state machines behind one internal API covering card, netbanking, UPI, OTP, and refund flows — adopted by brands including Domino's and Dunkin'.

EDUCATION

Bachelor of Technology in Information Technology 07/2018 – 05/2022 Ranchi, Jharkhand
Birla Institute of Technology, Mesra | CGPA 8.2 / 10.0

AWARDS & ACHIEVEMENTS

- Chairperson Award** — for significantly improving Rider Incentive payment accuracy while reducing operational costs.
- Personal Gold Award** — for contributions to developing the Dynamic Loyalty Service.
- Winner, company-wide hackathon (2026)** — among 35 teams, with the ML-based delivery time prediction model now running in production.
- Podium finish (top 3), company-wide hackathon (2024)** — among 38 teams, with the 'Common Cart' idea.

INTERESTS

Chess · Cricket · Table Tennis